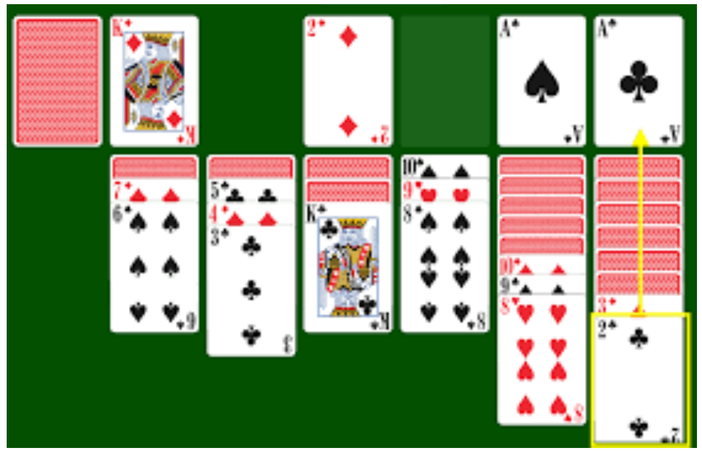
Montecarlo Simulations

An introduction

m colarieti tosti, 20 march 2023

Montecarlo: using random numbers and statistics to determine definite quantities



It was (allegedly) invented by Stanislaw Ulam as a way to determine the probabilities in a complex combinatorics problem:

> ***Given a game strategy, what is the probability of winning at solitaire?***

Montecarlo solution:

repeat N times

1. Shuffle the deck (i.e. order randomly the 52 cards)
2. Play the game (you need a strategy for placing the cards handed from the deck)
3. Check if you win or not
4. Update the N_wins if you won

Prob. of winning ≈ N_wins/N if N is large enough

Montecarlo can be used for many disparate things: Calculating light distribution in a scene



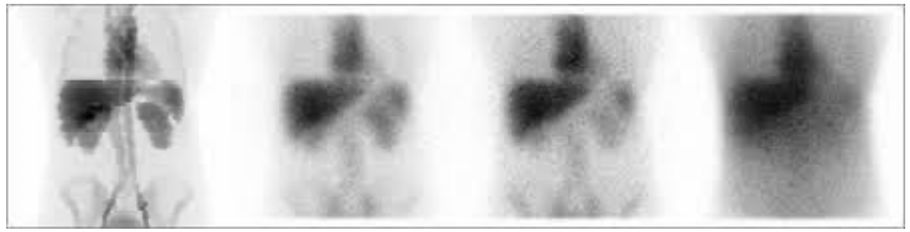


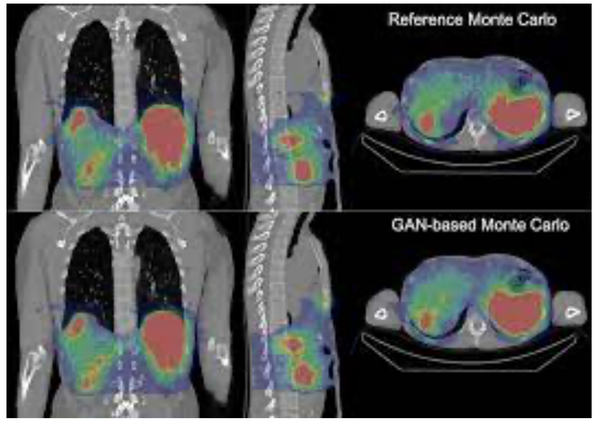
Generate many light rays in random directions => score where they end up => the more the brighter

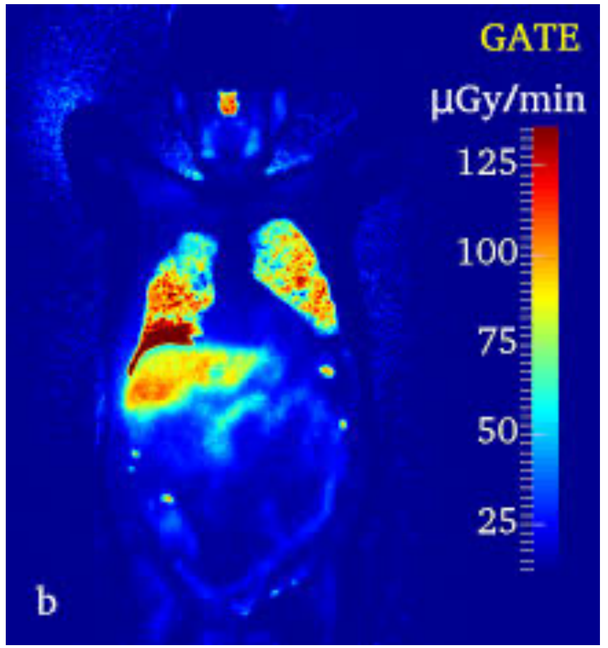
Here you can find a much more detailed explanation:

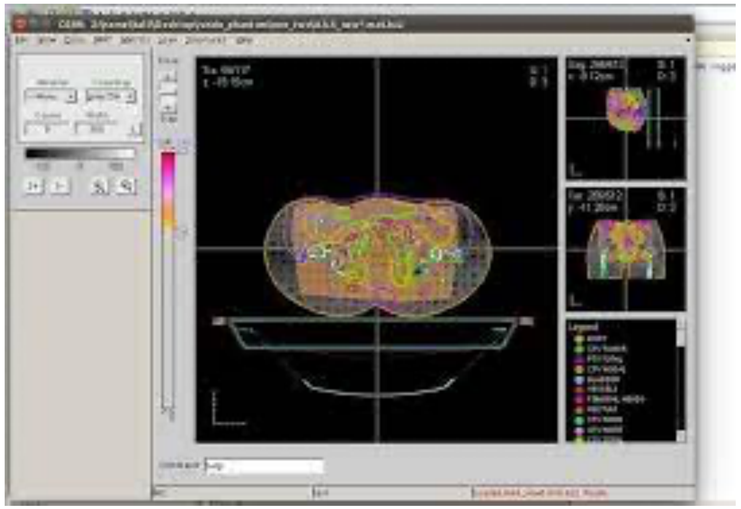
https://ocw.mit.edu/courses/electrical-engineering-and-computer-science/6-837-computer-graphics-fall-2012/lecture-notes/MIT6_837F12_Lec18.pdf

… and it can be used for simulating transport of ionising radiation through matter and simulate images and dose distributions with great accuracy:



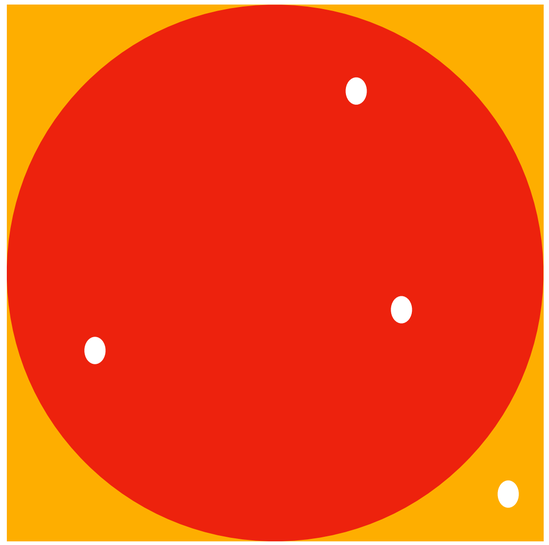






Montecarlo: Understanding the principles with simple examples

Let us pretend that I want to estimate π and I think of the following fact: $A_{disk}/A_{square} = \pi/4$



If I imagine to throw darts at this "target" randomly (that is without aiming at hitting nor missing the red disk) I should have:

$4N_{darts\ in\ disk}/N_{darts\ thrown} \approx \pi$. (If I throw enough darts)

OK let's shoot!

1 dart thrown, 1 dart in => $\pi \approx 4$

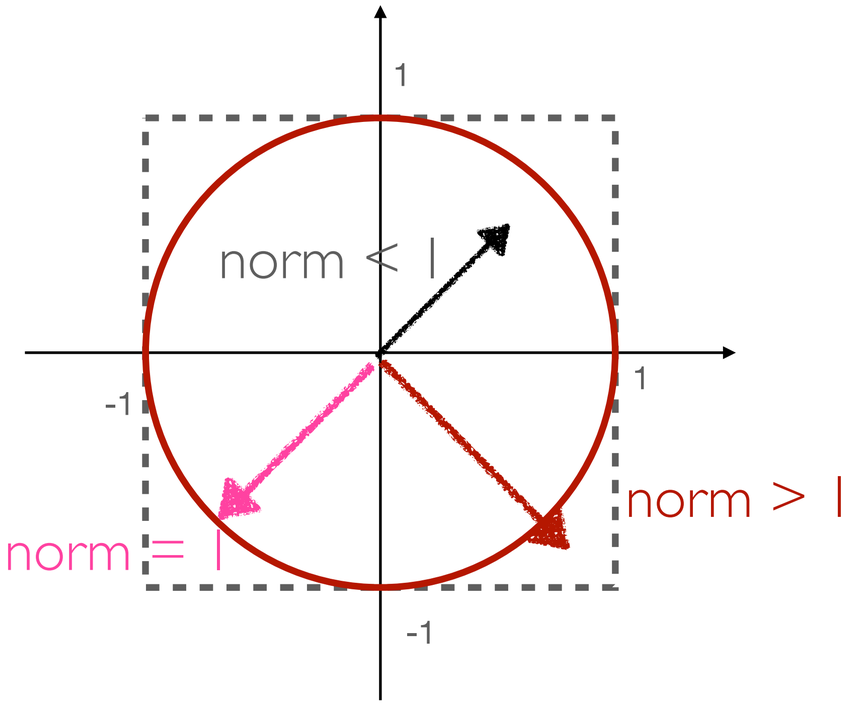
2 darts thrown, 2 darts in => $\pi \approx 4$

3 darts thrown, 2 darts in => $\pi \approx 2{,}67$

4 darts thrown, 3 darts in => $\pi \approx 3$

…. let's write a code!

Montecarlo: Understanding the principles with simple examples (cont)



Code idea:

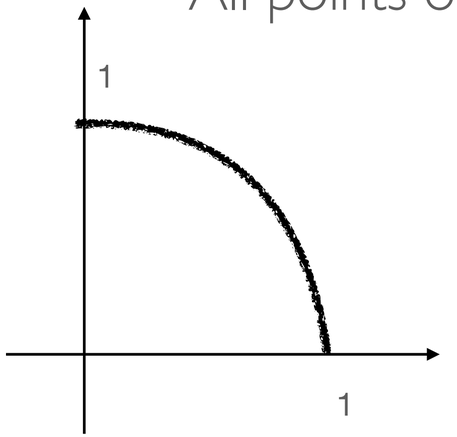
- sample x-coordinate randomly between -1 and 1
- sample y-coordinate randomly between -1 and 1
- calculate the L2-norm (x,y)
- count the times the norm is lower or equal to 1
- divide by the number of attempts
- multiply by 4 to get π estimate

Wait a second! How many times should I do this for? When is N large enough?

Wait a second! How do I know that I will get the right answer?

Integrating by means of MC-simulations

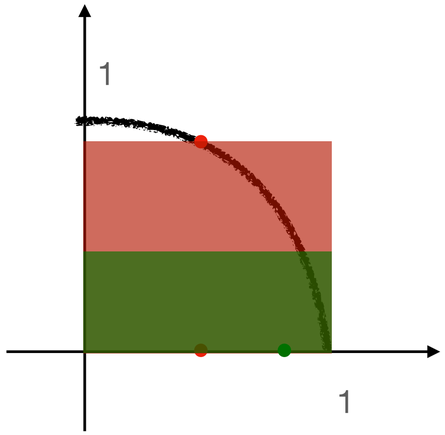
All points on the curve have distance 1 from origo. Do you know the function describing it?



$f(x) = \sqrt{1 - x^2}$ So, can you guess the value of the integral: $\int_0^1 \sqrt{1 - x^2} dx = ?$

If you run a Montecarlo in your head, by now, you should guess it to be π/4 (or maybe you simply realise it is 1/4 of the area of a disk with radius 1)

Let us now try another MC algorithm for finding the integral (let's suppose we do not know anything about geometry, for example)



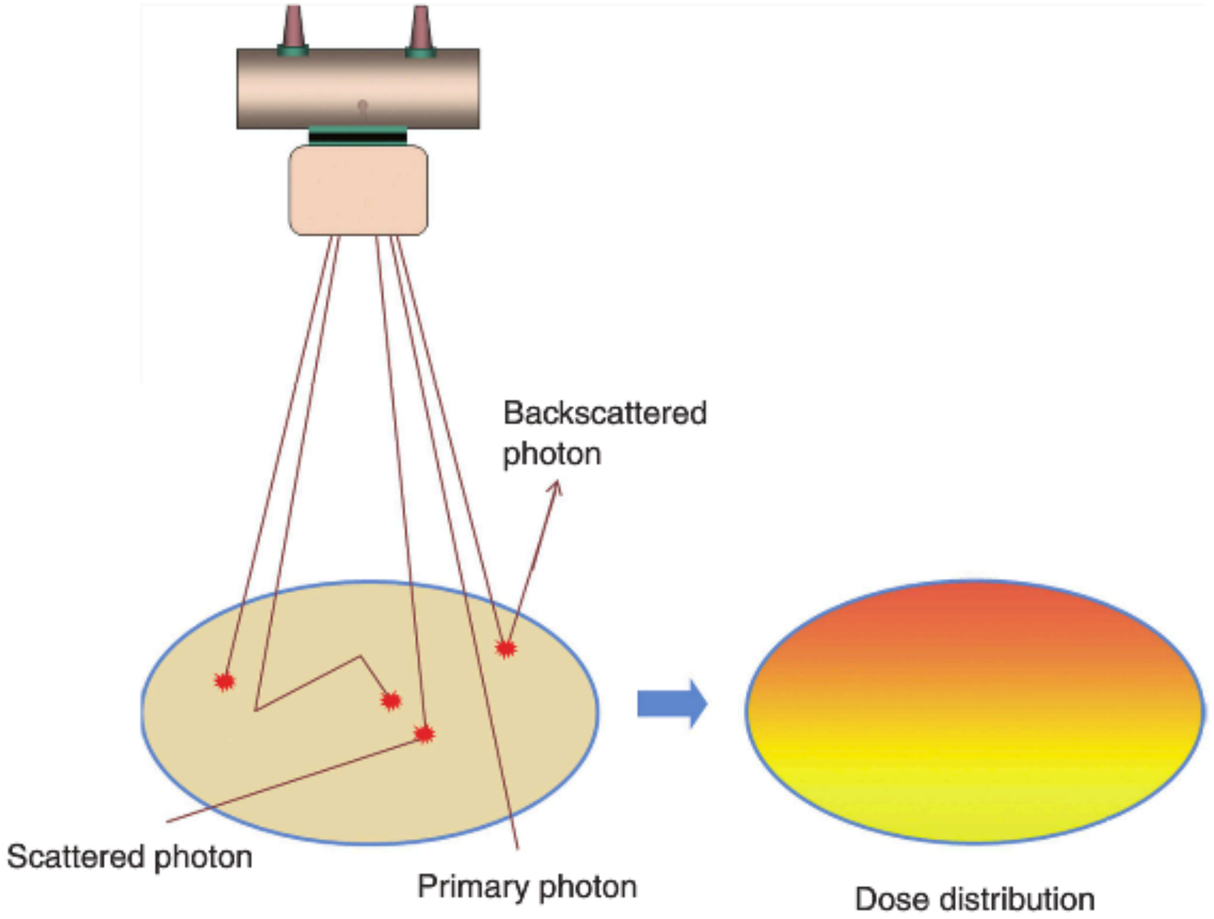
1. generate a random number, r, between 0 and 1
2. calculate f(r)
3. estimate your integral as f(r) (height) times 1 (base)
4. repeat averaging: integral = sum of all rectangles / number of rectangles

Integral ≈ $\dfrac{\text{▮} + \text{▬}}{2}$

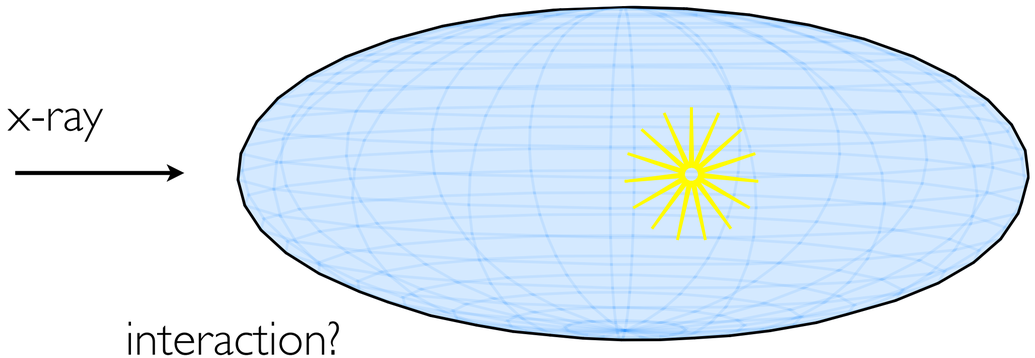
MC basic caveats

- Convergence
- Convergence to right answer?
  - bad model
  - bugs
- pseudorandom

Montecarlo: Understanding the principles with less simple examples



Montecarlo: Understanding the principles with less simple examples



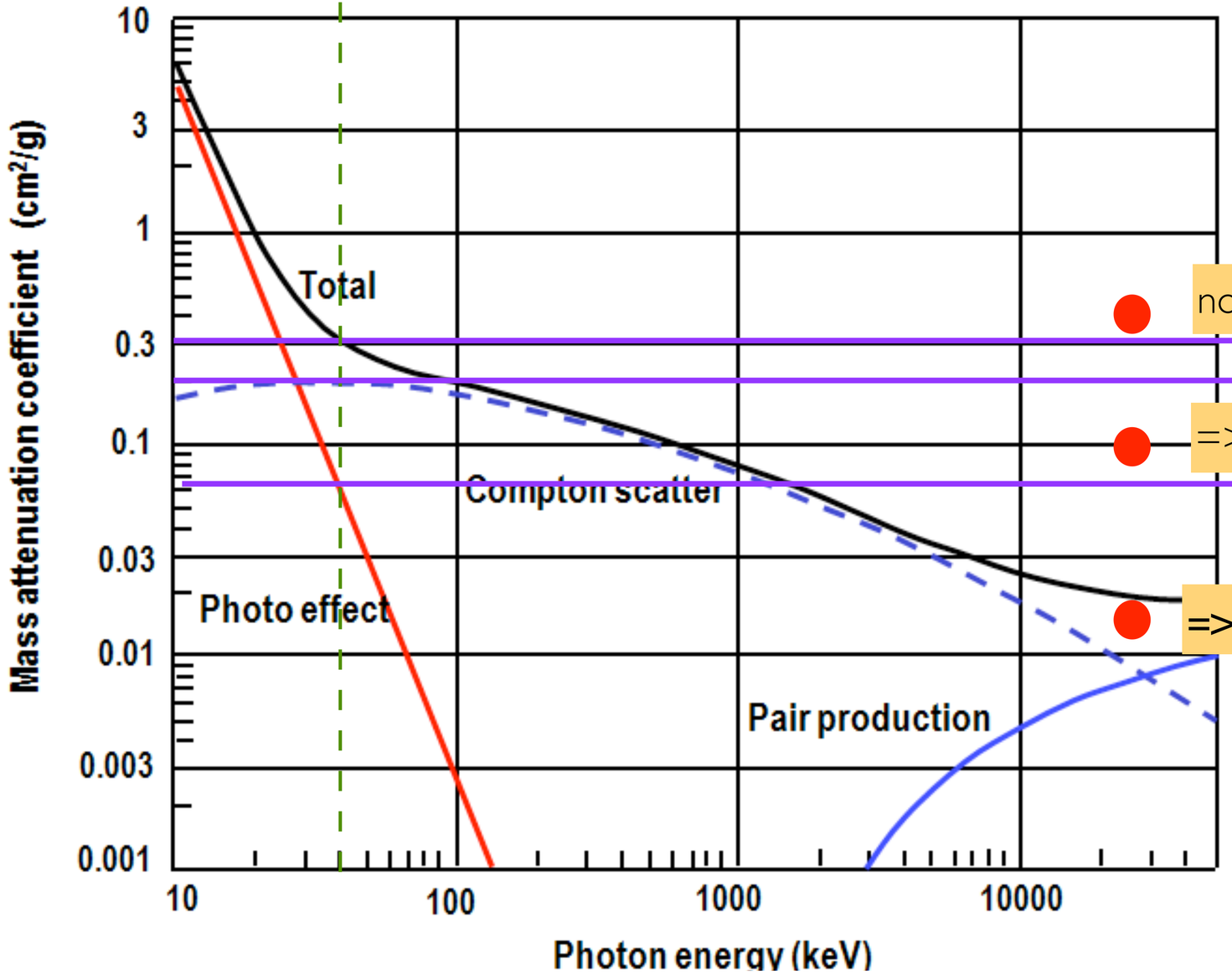


heads -> no interaction

what interaction?

heads -> PE

More realistic: γ@40keV



generate random number

nothing!

=>Compton!

=> PE!

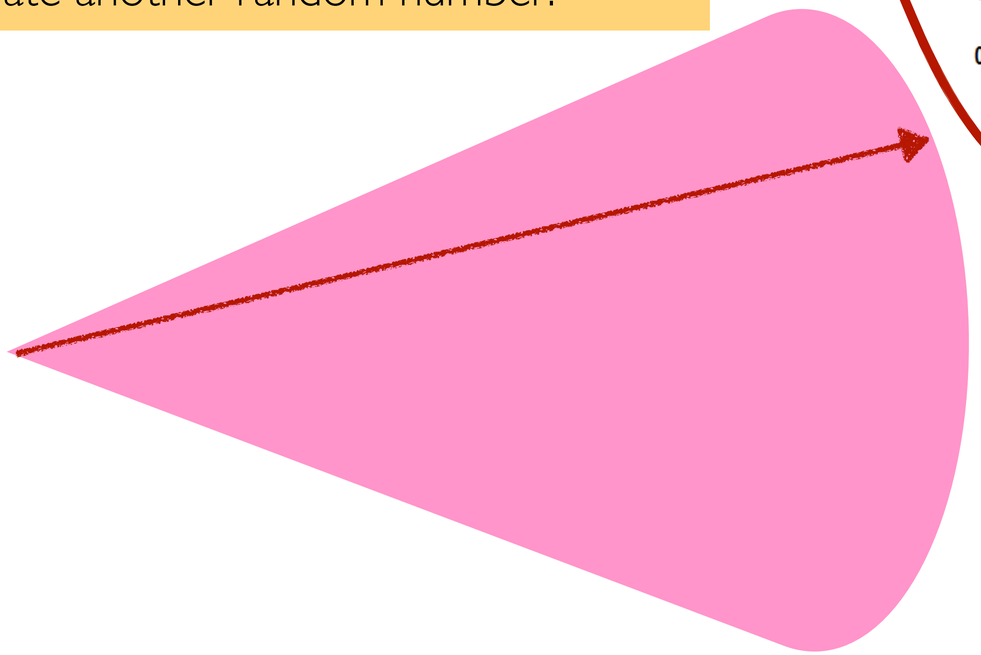
...but attenuation strongly depends on photon energy!

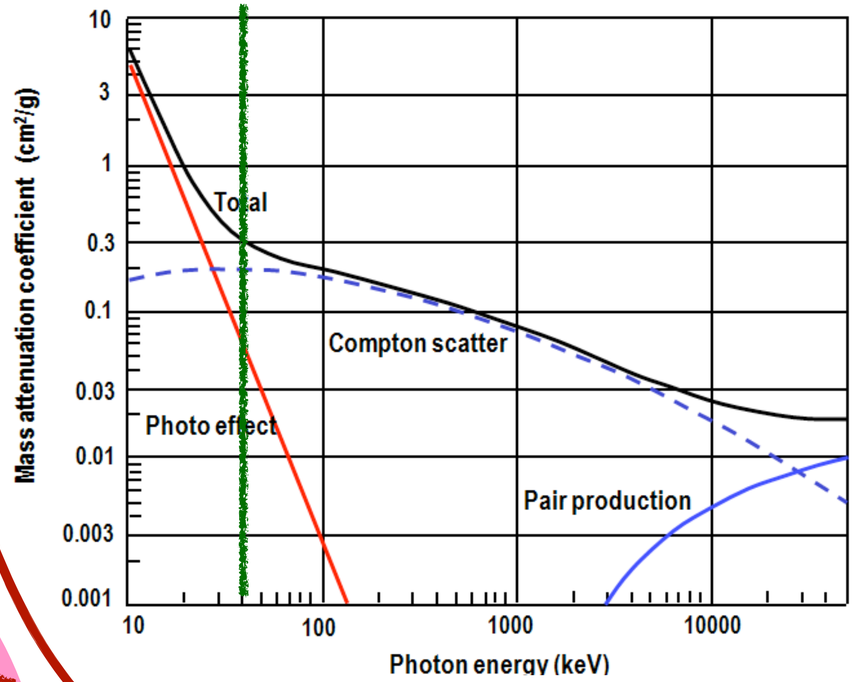
after Compton E must be updated!

generate another random number!

in general (3D) E does not fully specify direction

generate another random number!





Note that, in this case, the sampling distribution will not be uniform but will follow Klein-Nishina

… and how long free flight between interactions?

The probability for 1 foton of surviving for time t and then interact at t:

$p(t)dt = e^{-wt}wdt$, where $w = M\sigma c$, with M number of atoms per unit volume, σ cross-section for one atom and c speed of light in the material

Considering a time interval 0-t_c, the cumulative distribution function is:

$$\mathcal{P}(t_c) = \int_0^{t_c} we^{-wt}dt$$

We want to have $P(r) = P(t_c)$ with r random number. (For uniform random distr. simply $P(r) = r$.)
So we get:

$$\mathcal{P}(t_c) = \int_0^{t_c} we^{-wt}dt = 1 - e^{wt_c} = r$$

and inverting:

$$t_c = -\frac{1}{w}\ln(1 - r)$$

All together now!

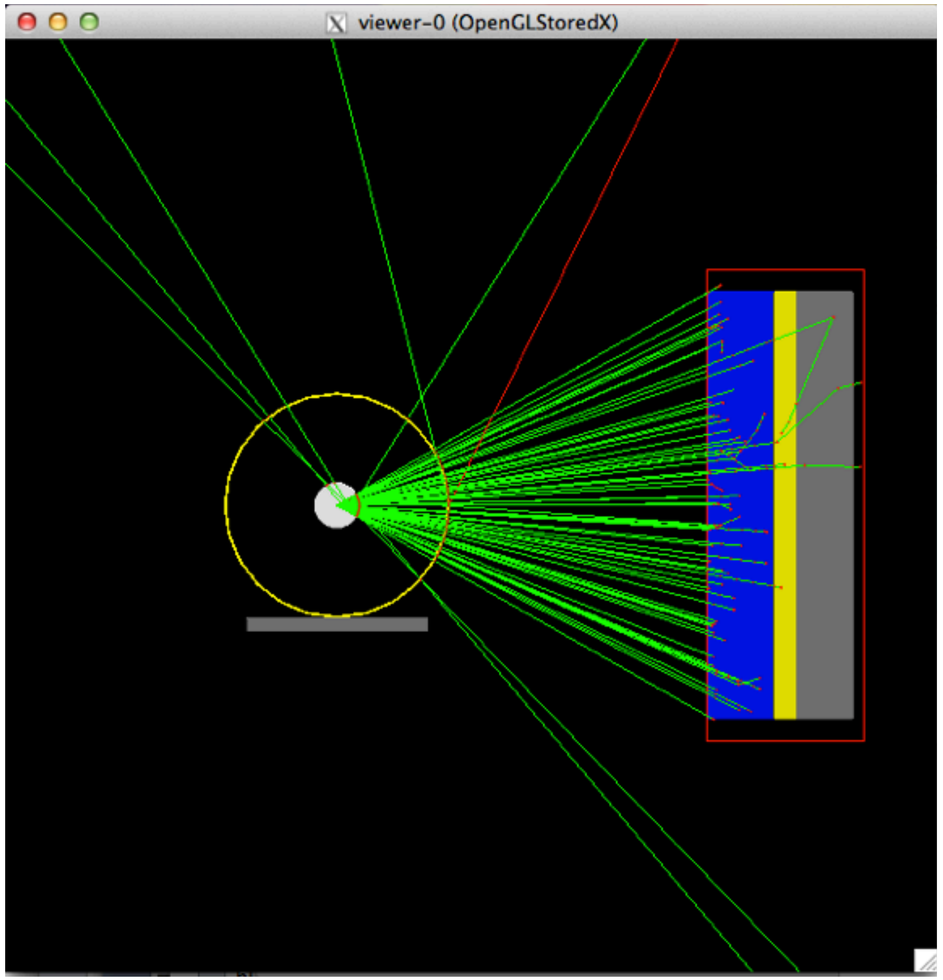


- initial E, direction
- generate random number r_1
- free-flight time $t = 1/w \cdot \ln(r_1)$
- propagate t, update position
- generate random number r_2 identify interaction
- generate 2 random numbers for new direction.
- Update energy, direction

4 random numbers!!!!

You do not need to write your own MC code!

GATE (http://www.opengatecollaboration.org)
Based on GEANT (CERN 30+ years of development and debugging)



and there are many, many more

Lab exercise:

Choose one of the following 3 tasks:

Evaluate the following integral using the "darts" and the "rectangles" method.
Study the evolution of your estimate and its variance as a function of number of samples
Write a short report explaining your codes and with your results presented in graphs

$$\int_0^{7,5} \frac{x^{x+1}}{\ln(x+3)e^x} dx$$

Write a pseudo code for a MC simulation for x-ray imaging. Write a short report explaining your code and discuss how you could assess the quality of your simulation.

Write a pseudo code for a MC simulation for evaluating the dose received by a patient undergoing x-ray imaging. Write a short report explaining your code and discuss how you could assess the quality of your simulation.